Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Slope of the line through (1; 2) and (3; 8)?

2. Expand $(2 + \quad h)^2 - 4$.

3. Simplify $\dfrac{4h + h^2}{h}$ if $h \neq 0$.

4. Calculate $\sqrt{0}$ and $\sqrt{0{,}01}$.

5. Calculate $|-3|$ and $|2|$.

6. A line with slope 2 passes through (1; 5). Equation?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Explain to a classmate why we simplify the rate before letting h tend to 0. Distinguish average and instantaneous speed.

Investigate before formalising

For $f(x) = x^2$, vary h, positive and negative, around 0, keeping point A at x = 1.
Draw three secants; conjecture a limiting line, then determine it without reading the graph.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min.
Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.